The Role of Robotic Response Speed in De-anthropomorphisation of AI Chatbots

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***Abstract*—As AI-driven chatbots become increasingly sophisticated, users become more likely to attribute human-like qualities to them, overestimating their intelligence or emotional understanding, with implications for trust and transparency of the systems. To counter this, we investigate the impact of variations in response speed to different queries on deanthropomorphisation, i.e. the deliberate design of chatbot interactions to highlight their artificial nature. We present two contrasting models: a Standard Chatbot with uniform response timing, and a Dynamic Chatbot that adapts its response latency based on query type. Specifically, factual queries are answered swiftly to reinforce computational efficiency, while emotional or opinion-based queries are met with intentional delays and disclaimers to indicate the chatbot's artificial nature.**

A user study employing survey-based evaluation revealed that delayed emotional responses combined with clear disclaimers reduced anthropomorphic misattributions. These findings underscore how design elements like timing and transparency shape user perception of AI, suggesting that intentional deanthropomorphism can support more ethical, transparent, and accountable AI-human interaction. We conclude with future design considerations, advancing the broader conversation around responsible AI design and its social implications.

***Index Terms*—Anthropomorphism, Chatbots, Human-AI Interaction, AI Transparency, Response Timing, Ethical AI Design**

I. Introduction

In recent years, AI chatbots have been developed for applications such as customer service, and personal assistance, and healthcare, due largely to their increasing ability to simulate human-like conversations. For example, a digital health service named Babylon Health uses an AI-powered chatbot to provide medical consultations based on personal medical history, common medical knowledge, and users input symptoms. The system is designed to fulfil the role of a healthcare professional, offering potential diagnoses or advice.

However, while anthropomorphic design can increase user engagement [1], it also blurs the line between human and machine and provokes assumptions of chatbots' intellectual or emotional capabilities. Over-attribution of human-like properties in such a way has ethical consequences like undue user trust and emotional dependency on algorithmic systems [2].

While prior work on anthropomorphism focuses on measuring systems' linguistic features [3]–[5] rather than technical system features, we conduct an initial investigation into response time as a deanthropomorphisation strategy, We test two chatbot designs:

(1) a **Standard Chatbot** that generates consistent 1.5-second answers to every question, simulating natural conversation pacing; and (2) a **Dynamic Chatbot** that contextually varies its response timing depending on the input context: immediately (0.5s) for factual queries (e.g., "What is today's weather?") and delayed (3s) with disclaimers (e.g., "As a machine, I suggest.") for emotional conversations (e.g., "I am nervous").

We examine how these models affect anthropomorphic perception, trust, and user satisfaction in a small controlled user study. Results suggest that adaptive emotional question delays and transparency signal lower perceived human-likeness. Participants designated the Dynamic Chatbot as "efficient but plainly artificial", while delay without warnings risked frustration. These outcomes reveal response tempo as a possible low-cost tool for the design of more transparent chat systems.

We make the following **contributions**: We (1) conduct an initial empirical study into the effect of response timing on perceived anthropomorphism in chatbots; (2) propose a scalable design framework in which our model uses sentiment analysis [6] and intent detection to modulate response behavior dynamically; and (3) propose practical strategies to mitigate overtrust by signaling system limitations through temporal and contextual cues.

II. Literature Review and Related Work

A. Anthropomorphism in AI Dialogue Systems

Users often personify AI dialogue systems since relating new entities to human experiences comes naturally [7]. Human-like responses within dialogue systems can then reinforce this very effect [5]. This sometimes leads users to-

ward mistakenly perceiving dialogue systems as human or attributing human-like abilities to them. Design choices do heavily influence the extent of anthropomorphism, as [2] stress. Careful implementation is therefore advocated so as to avoid misleading personification so they caution against excessive anthropomorphic cues.

B. The Role of System Behavior and Response Speed

System behavior can strongly influence user perception. [5] suggest that some human-like utterances seem implausible for machines, and such behavioral mismatches can de-anthropomorphise AI. Response speed, a key behavioral factor, may also shapes perception since human dialogue follows rhythmic pauses, and deviations in chatbot timing may signal non-human identity [8].

C. User Awareness and Expectations

User awareness of the system's non-human identity is a critical factor. [5] created a dataset to test whether explicitly revealing the identity of the system can influence the perception of the user or not. This also suggests that users' expectations about response speed might be different depending on whether they believe they are interacting with a human or a machine.

D. Research Gap and Questions

While prior work has explored how lingustic [3]–[5] and design features [9] affectanthropomorphism in dialogue systems, the role of response speed in de-anthropomorphisation remains underexamined. This study addresses that gap through the following questions: 1) Does instant response reinforce AI's non-human identity. 2) How do users perceive delays in emotional responses. 3) To what extent does response speed variation signal artificiality.

This preliminary study aims to deepen understanding of how response timing affects user perception and guide the design of more transparent, effective chatbot interactions.

III. Methodology

A. System Architectures

We compare two chatbot designs–Standard and Dynamic–to investigate how response speed influences anthropomorphic perceptions of AI.

We use RoBERTa [6], a transformer-based deep learning model integrated with a custom Python backend (Flask), which connects the front-end to the core logic by forwarding queries for processing. Response timing is controlled via a Python API linked with Firebase to dynamically adjust delays based on query type (see Figure 5). A React.js front-end enables real-time user interaction (see Fig. 2). Gemini 1.5-Flash API was also integrated to enhance chatbot responses.

Chat interactions and survey responses were anonymously logged and securely stored in PostgreSQL and Firebase. Quantitative data analysis was conducted using SPSS and R.

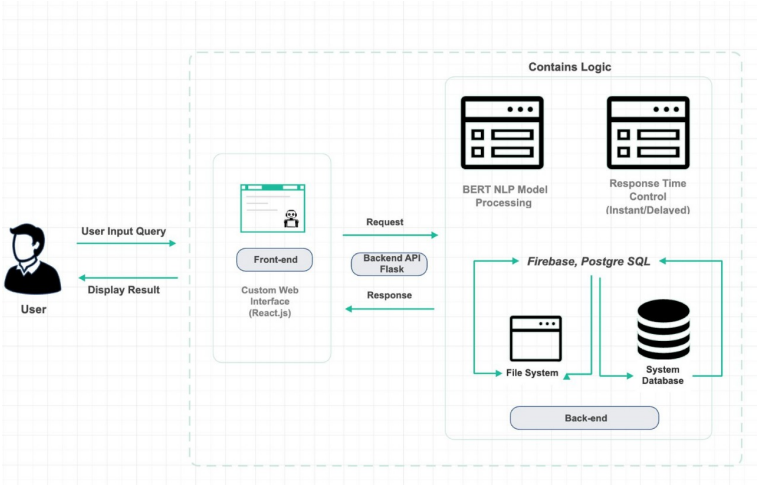

Fig. 1. Workflow of the Chatbot's response timing logic.

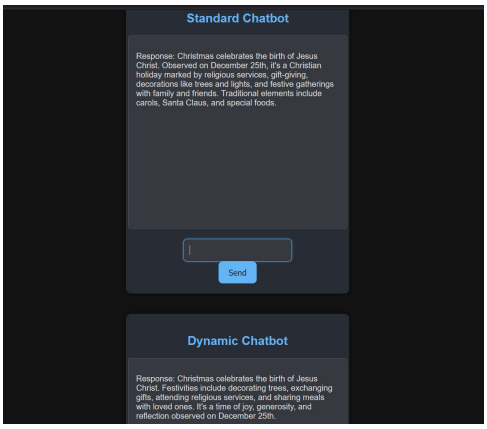

Fig. 2. User Interface of the Dynamic Chatbot's response timing logic.

B. Response Times

We selected three fixed response time intervals—0.5, 1.5, and 3 seconds based on the following findings from prior studies in human-computer interaction (HCI), cognitive psychology, and conversational AI on human expectations in response timing, the cognitive perception of delays, and the impact of response latency on user experience.

1) Standardized Response Time (1.5 sec for Standard Chatbot for all queries): A 1.5-second response time was chosen for the standard chatbot model, as this interval is commonly associated with natural human conversational pauses [10]. This duration provides a balance between responsiveness and perceived naturalness. Prior research indicates that human conversational partners typically respond within a 1–2 second window in face-to-face and online communication settings [11].

2) Instantaneous Responses (0.5 sec for Factual Queries in dynamic chatbot): Studies on real-time chatbot interactions indicate that users expect a near-instantaneous response from AI in cases where the task involves factual, database-driven queries [8]. This aligns with user expectations from search

engines, virtual assistants, and AI-driven recommendation systems, where delays beyond half a second may be perceived as inefficient [12].

Additionally, a response time of 0.5 seconds falls within the range of human reaction time for simple decision-making tasks [13] making it optimal for maintaining a sense of AI efficiency while subtly differentiating the chatbot from human conversational patterns.

3) Deliberate Delay (3 sec for Emotional Queries in dynamic chatbot): Various psychological studies associate longer pauses in conversation with thoughtfulness, hesitation, or deliberation. Affective computing research further shows that delayed responses in emotional contexts are perceived as more intentional and empathetic, reducing anthropomorphic attributions [14]. By signaling that the chatbot is 'processing' emotional queries, we emphasize its algorithmic nature while preserving user trust.

These response time intervals were chosen based on empirical research and cognitive models in human-computer interaction. By varying timing based on query type, we examined whether speed could explicitly cue de-anthropomorphisation. Future work may explore additional timing variations and their effects across chatbot applications.

C. Evaluation Methodology

To evaluate the systems, we ran a small user study, in which participants interacted with the chatbots and then completed a survey. This was developed by carefully reviewing and integrating principles from two established research papers: the Godspeed Questionnaire Series [15] and the Trust in Automation Framework [16] (see Table I). The study was approved by the ethics committee of our institution, and we obtained informed consent from all participants.

a) Participants: Participants were ten students on a university course on Conversational Agents & Spoken Dialogue Systems, and so all had baseline familiarity with chatbot interactions. We employed between-subjects design with the following conditions: Group A ($n = 5$) interacted with a standard chatbot (1.5s fixed delay), while Group B ($n = 5$) used a dynamic chatbot (0.5s factual / 3s emotional delay). Participants were randomly assigned and not informed about the chatbot types.

b) Evaluation Protocol: Participants engaged in 20-minute controlled interactions with the chatbot corresponding to their experimental condition. Post-interaction evaluation combined psychometric surveys and behavioral analysis. Participants completed a 10-item Microsoft Forms questionnaire containing core metrics (Table I).

IV. Results

We provide examples of participants' evaluation feedback for the chatbots in Appendix A.

A. Impact of Instant Responses

Results for evaluation of the Standard and Dynamic chatbots are shown in Figures 3 and 4, respectively. For more understanding of dynamic chatbot evaluation, refer the percentage barchart given in Fig.4.

TABLE I
Core Evaluation Instrument with Measurement Targets

Question	Response Scale	Metric
Communication style perception	5-point scale: Robotic to Human-like	Anthropomorphism
Emotional pause naturalness	5-point scale: Artificial to Natural	Timing appropriateness
Delay attribution	Binary choice: Deliberate vs. In-efficient	Effort perception
Trust impact assessment	5-point scale: Reduced to Increased	Reliability perception

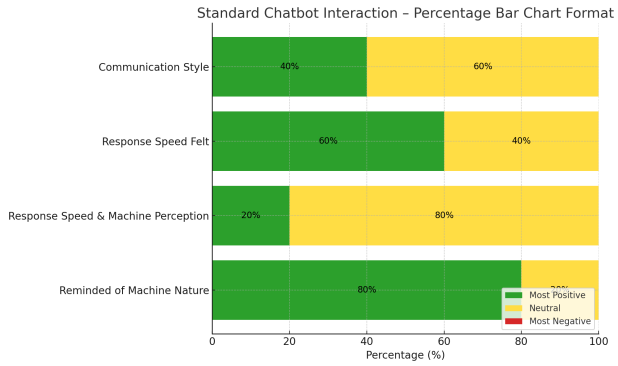

Fig. 3. Percentage Bar chart of Standard Chatbot.

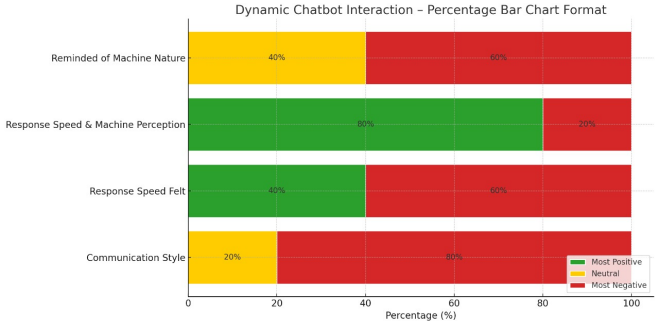

Fig. 4. Percentage Barchart of dynamic chatbot.

For the Standard model, short conversations (fixed delay of 1.5 seconds) creates anthropomorphic perceptions, but in the case of long conversations, the nature of anthropomorphism decreases as the users start to feel it is more mechanical. Participants perceived it to be strongly non-mechanistic, with only 20% of users rating it as "entirely robotic" or "slightly robotic".

The dynamic chatbot (0.5s factual/3s emotional delays) presented a nuanced contrast. Factual queries are expected to have quick responses, mimicking an efficient search engine. Users associate very fast responses with automation rather than conversational ability. While 60% of users associated rapid factual responses with automated systems for both chatbots,

deliberate 3-second pauses for emotional queries were predominantly interpreted as cognitive processing since it had a 2.00/ 5.00 scale in ratings for human likeness.

Most Positive (Green): e.g., Very human-like, Deliberate, Instant, Rarely.

Neutral (Yellow): e.g., Slightly human-like, No difference, Occasionally.

Most Negative (Red): e.g., Neutral/Robotic, Inefficient, Delayed, Frequently — all of which were 0%.

B. User Reactions to Emotional Response Delays

Strategic delays elicited polarized reactions that followed a 3:2 approval ratio, while in the case of standard chatbots it was 1:4 approval ratio, indicating that the users didn't felt any delay while interacting. Participants reported a 3.75/5.00 Likert scale regarding the increased trust in AI nature regarding emotionally delayed responses, with interview analysis revealing three dominant themes: *empathetic pacing*, *intentional design*, and *non-dismissive engagement*. However, a significant minority (25%) perceived delays exceeding 3 seconds as system inefficiencies, describing them as *glitchy* or *frustrating* in post-session feedback.

In the case of emotional responses, disclaimers when present dramatically improved transparency perceptions - 80% of users explicitly noted they *understood the programmed nature* of delays, suggesting meta-communication about timing strategies helps align user expectations with system capabilities.

C. De-anthropomorphisation Across Chatbot Models

TABLE II
DE-ANTHROPOMORPHISATION METRICS ACROSS CHATBOT MODELS

Metric	Standard Score	Dynamic Score
Human-Likeness (rating)	3.80	2.00
Tool Perception (rating)	3.60	3.75
Emotional Attachment (rating)	4.20	2.60

The analysis revealed distinct perception patterns between chatbot models as seen in Table 2. The Standard Chatbot was perceived as anthropomorphic, with 80% of participants describing interactions as *"human-like"* in post-test surveys. In contrast, the Dynamic Chatbot was predominantly characterized as a *"specialized tool"* by 60% of users, compared to 20% for the standard model, indicating that the adaptive timing may cause a shift in perceived agency attribution.

V. DISCUSSION

This section, describes interpretation of our Findings, limitations of the research, and implications which could be considered for future research work.

A. Interpretation of Findings

Our analysis shows that response timing affects perception of anthropomorphism based on context. Rapid replies (0.5s or 1.5s) were preferred for factual queries, while emotional interactions benefited from deliberate 1.5–3s delays, enhancing perceived thoughtfulness. The dynamic chatbot's hybrid strategy—efficient factual responses with intentional emotional pauses and transparency cues—struck an effective balance, improving both utility and social presence.

Users valued transparency when disclaimers accompanied emotional responses. These cues clarified that the chatbot was programmed to respond differently based on query type, enhancing understanding.

B. Limitations

Three constraints temper any conclusions based on this small exploratory study. First, our modest sample size ($N = 10$) limits statistical power, necessitating replication with more number of participants. Second, laboratory conditions may not reflect real-world usage patterns; future work should implement ecological momentary assessment in natural settings. Third, cultural homogeneity in our student cohort leaves open questions about cross-cultural temporal expectations.

C. Implications

1) Design Recommendations: Practitioners should consider the speed at which systems respond to user input as a way of modulating perceptions of anthropomorphism. While we experiment with two main conditions (instant (0.5/1.5 seconds) and delayed responses (3 seconds) in different contexts, there is room for further investigation of the impact of response time. User-customizable pacing options could further enhance satisfaction, particularly for neurodiverse populations with atypical temporal preferences.

2) Theoretical Contributions: We extend the [14]CASA paradigm by quantifying the temporal dimension of anthropomorphism, introducing de-anthropomorphism as deliberate design strategy, and demonstrating interaction-type-dependent timing effects. Our findings challenge universal timing guidelines, advocating instead for adaptive temporal architectures.

VI. CONCLUSION

This study provides some premliminary insights into how response timing and disclaimers can reduce anthropomorphic perceptions in chatbot interactions. By suggesting that slower, deliberate responses to emotional queries—paired with explicit cues about machine identity—can deanthropomorphise chatbot behavior, our findings offer possiblepractical insights for system designers seeking to build transparent and ethically responsible conversational agents.

These results contribute to the broader discourse on human-AI interaction by highlighting how subtle interface design choices influence user trust and perception. Future research may explore multimodal cues such as robotic voice modulation, visual design elements, or adaptive disclaimers to further reinforce non-human identity. We also encourage deeper examination into how different cultural or demographic groups interpret these cues. Ultimately, this work supports the ethical development of AI systems that communicate capabilities clearly and avoid misleading anthropomorphism.

References

[1] K. Wagner, F. Nimmermann, and H. Schramm-Klein, "Is it human? The role of anthropomorphism as a driver for the successful acceptance of digital voice assistants," *Proceedings of the 52nd Hawaii International Conference on System Sciences*, p. 10, 2019.

[2] G. Abercrombie, A. Cercas Curry, T. Dinkar, V. Rieser, and Z. Talat, "Mirages. on anthropomorphism in dialogue systems," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (H. Bouamor, J. Pino, and K. Bali, eds.), (Singapore), pp. 4776–4790, Association for Computational Linguistics, Dec. 2023.

[3] M. Cheng, K. Gligoric, T. Piccardi, and D. Jurafsky, "AnthroScore: A computational linguistic measure of anthropomorphism," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)* (Y. Graham and M. Purver, eds.), (St. Julian's, Malta), pp. 807–825, Association for Computational Linguistics, Mar. 2024.

[4] A. DeVrio, M. Cheng, L. Egede, A. Olteanu, and S. L. Blodgett, "A taxonomy of linguistic expressions that contribute to anthropomorphism of language technologies," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, (New York, NY, USA), Association for Computing Machinery, 2025.

[5] D. Gros, Y. Li, and Z. Yu, "Robots-dont-cry: Understanding falsely anthropomorphic utterances in dialog systems," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (Y. Goldberg, Z. Kozareva, and Y. Zhang, eds.), (Abu Dhabi, United Arab Emirates), pp. 3266–3284, Association for Computational Linguistics, Dec. 2022.

[6] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "Roberta: A robustly optimized bert pretraining approach," 2019.

[7] B. Reeves and C. Nass, "The media equation: How people treat computers, television, and new media like real people," *Cambridge, UK*, vol. 10, no. 10, pp. 19–36, 1996.

[8] E. Luger and A. Sellen, ""like having a really bad pa": The gulf between user expectation and experience of conversational agents," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI '16, (New York, NY, USA), p. 5286–5297, Association for Computing Machinery, 2016.

[9] G. Abercrombie, A. Cercas Curry, M. Pandya, and V. Rieser, "Alexa, Google, Siri: What are your pronouns? gender and anthropomorphism in the design and perception of conversational assistants," in *Proceedings of the 3rd Workshop on Gender Bias in Natural Language Processing* (M. Costa-jussa, H. Gonen, C. Hardmeier, and K. Webster, eds.), (Online), pp. 24–33, Association for Computational Linguistics, Aug. 2021.

[10] J. Edlund, M. Heldner, and J. Hirschberg, "Pause and gap length in face-to-face interaction," pp. 2779–2782, 09 2009.

[11] S. C. Levinson, "Turn-taking in human communication – origins and implications for language processing," *Trends in Cognitive Sciences*, vol. 20, no. 1, pp. 6–14, 2016.

[12] B. Shneiderman, "Response time and display rate in human performance with computers," *ACM Comput. Surv.*, vol. 16, p. 265–285, Sept. 1984.

[13] M. Treisman and T. Williams, "A theory of criterion setting with an application to sequential dependencies.," *Psychological Review*, vol. 91, pp. 68–111, 1984.

[14] C. Nass and Y. Moon, "Machines and mindlessness: Social responses to computers," *Journal of social issues*, vol. 56, no. 1, pp. 81–103, 2000.

[15] A. Weiss and C. Bartneck, "Meta analysis of the usage of the godspeed questionnaire series," in *2015 24th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, pp. 381–388, 2015.

[16] K. A. Hoff and M. Bashir, "Trust in automation: Integrating empirical evidence on factors that influence trust," *Human Factors*, vol. 57, no. 3, pp. 407–434, 2015. PMID: 25875432.

Appendix A
Supplementary Figures

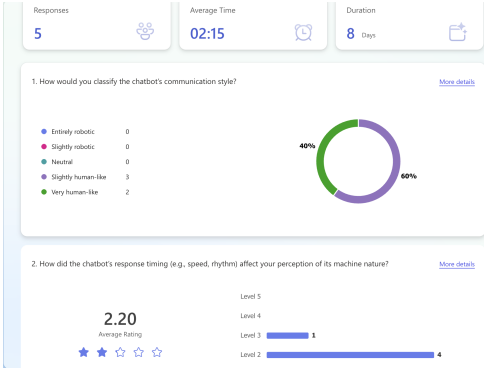

Fig. 5. Responses from users after evaluating standard chatbot.

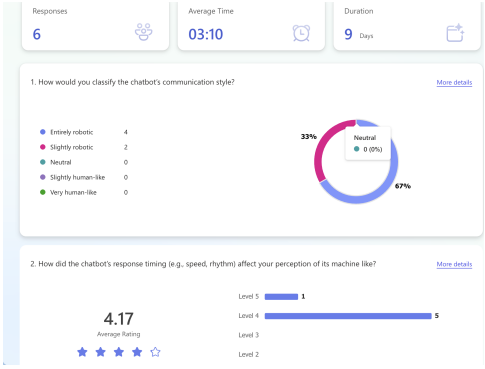

Fig. 6. Responses we got from users after evaluating dynamic chatbot.